

# Planetary Core Energy Maintenance via Solar Wind Absorption

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## PAPER\_1083: Planetary Core Energy Maintenance via Solar Wind Absorption

### Abstract

We derive the energy budget for planetary core maintenance through solar wind absorption. The core receives power  $P_{\text{wind}} = \eta_{\text{pen}} \cdot \Phi_{\text{sw}}(d) \cdot \pi R_{\text{mag}}^2$  from wind kinetic flux, which must balance Um maintenance cost  $P_{U_m} = \mu_{\text{core}}^2 \omega_{\text{core}} / \tau_{U_m}$  and radiative losses  $P_{\text{rad}} = \sigma T_{\text{core}}^4 \cdot 4\pi R_{\text{core}}^2$ . The net energy rate  $dE/dt = P_{\text{wind}} - P_{U_m} - P_{\text{rad}}$  determines whether a planet's core is in thermal equilibrium or depleting. The core SCm+UA is exclusively interactive with Ug3.

### §1 Wind Power Absorbed by Core

$$P_{\text{wind}} = \eta_{\text{pen}} \cdot (1 - \eta_{\text{liquid}}) \cdot \frac{1}{2} \rho_{\text{sw}}(d) \cdot v_{\text{sw}}^3 \cdot \pi R_{\text{mag}}^2$$

### §2 Um Maintenance Cost

$$P_{U_m} = \frac{\mu_{\text{core}}^2 \cdot \omega_{\text{core}}}{\tau_{U_m}}$$

where  $\mu_{\text{core}}$  is the planetary dipole moment,  $\omega_{\text{core}}$  the rotation rate, and  $\tau_{U_m} \sim 10^{15}$  s the magnetic diffusion timescale.

### §3 Radiative Loss

$$P_{\text{rad}} = \sigma T_{\text{core}}^4 \cdot 4\pi R_{\text{core}}^2$$

### §4 Core Energy Budget

$$\frac{dE_{\text{core}}}{dt} = P_{\text{wind}} - P_{U_m} - P_{\text{rad}}$$

$$\tau_{\text{depletion}} = \frac{E_{\text{core}}}{|dE/dt|}$$

Equilibrium condition:  $dE/dt \geq 0 \iff P_{\text{wind}} \geq P_{U_m} + P_{\text{rad}}$ .

## §5 Earth Example

| Parameter           | Value                               | Source               |
|---------------------|-------------------------------------|----------------------|
| $d$                 | 1.0 AU                              | Orbital distance     |
| $R_{\text{mag}}$    | $6.4 \times 10^7$ m                 | Earth magnetosphere  |
| $R_{\text{core}}$   | $3.486 \times 10^6$ m               | Inner+outer core     |
| $T_{\text{core}}$   | 5700 K                              | Core temperature     |
| $\mu_{\text{core}}$ | $8 \times 10^{22}$ A·m <sup>2</sup> | Geomagnetic dipole   |
| $P_{\text{wind}}$   | $\sim 10^{10}$ W                    | Wind power to core   |
| $P_{U_m}$           | $\sim 10^{15}$ W                    | Magnetic maintenance |
| $P_{\text{rad}}$    | $\sim 10^{19}$ W                    | Radiative loss       |
| $dE/dt$             | $< 0$                               | Net depletion        |

## §6 Ug3 Exclusivity

Core SCm+UA interacts exclusively with Ug3 (magnetic string rotation), not with Ug1 (dipole) or Ug2 (charge-reactivity). This means: - Core energy maintenance depends on Ug3 string coupling distance - Beyond the frost line, Ug3 coupling weakens and wind power dominates (PAPER\_1079) - Core dynamo longevity correlates with Ug3 string tension retention

## §7 Planetary Comparison

| Planet  | $P_{\text{wind}}$ | $P_{U_m}$        | $P_{\text{rad}}$ | Regime             |
|---------|-------------------|------------------|------------------|--------------------|
| Earth   | $\sim 10^{10}$ W  | $\sim 10^{15}$ W | $\sim 10^{19}$ W | Depleting (slowly) |
| Jupiter | $\sim 10^{14}$ W  | $\sim 10^{18}$ W | $\sim 10^{21}$ W | Depleting          |
| Neptune | $\sim 10^{11}$ W  | $\sim 10^{13}$ W | $\sim 10^{17}$ W | Depleting          |
| Mars    | $\sim 10^6$ W     | $\sim 0$ W       | $\sim 10^{16}$ W | Dead dynamo        |

## References

- PAPER\_1078: Solar Wind Flux Partitioning Budget
- PAPER\_1079: Frozen Planet Solar Wind Power
- PAPER\_413: Ug3CCWCWDifferentialRotationSCmPlanetaryCoreCalculator
- Star-Magic.txt: §5 Core Energy Maintenance